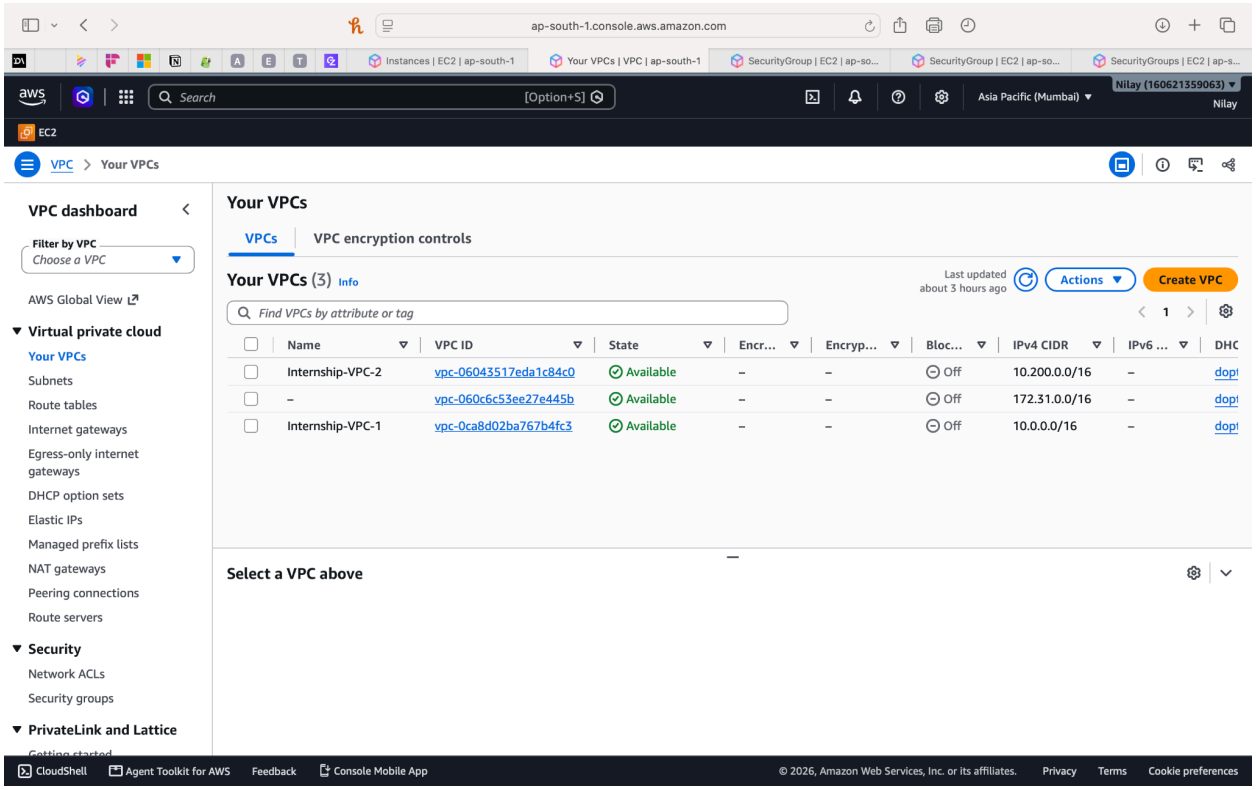
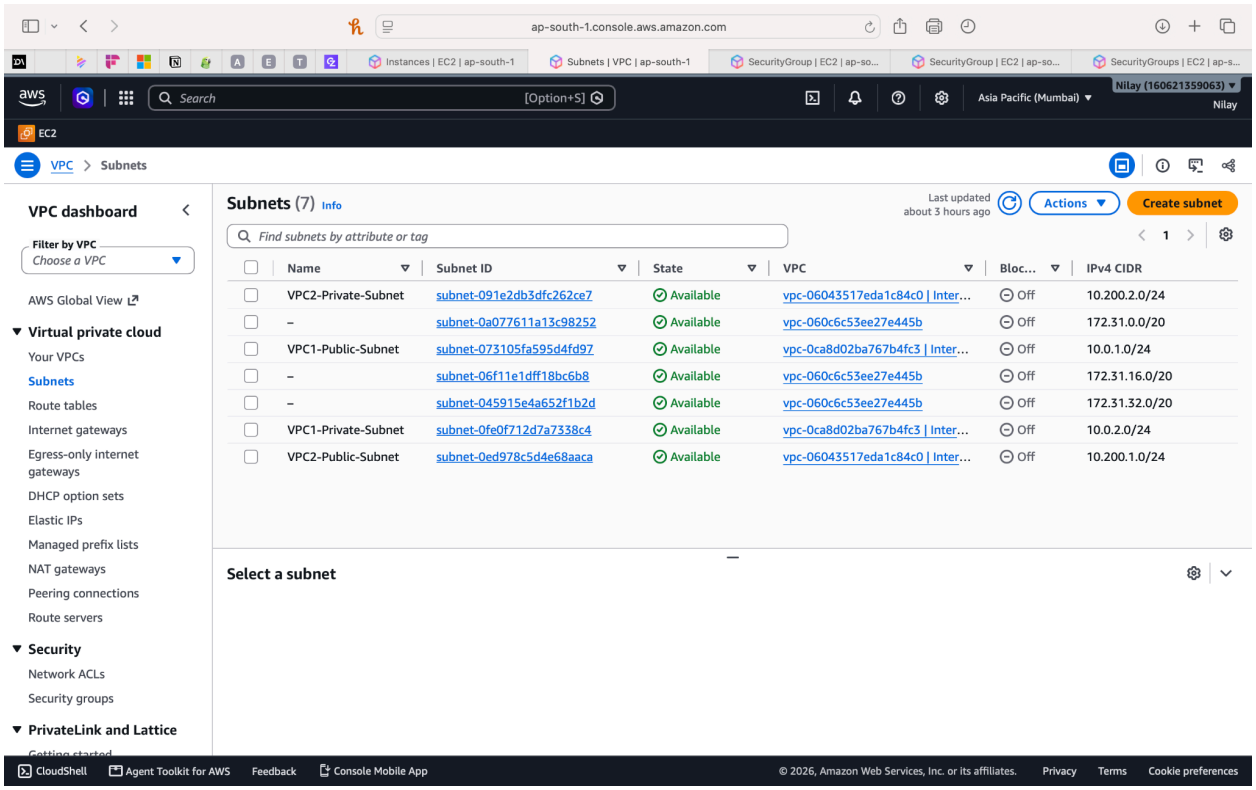
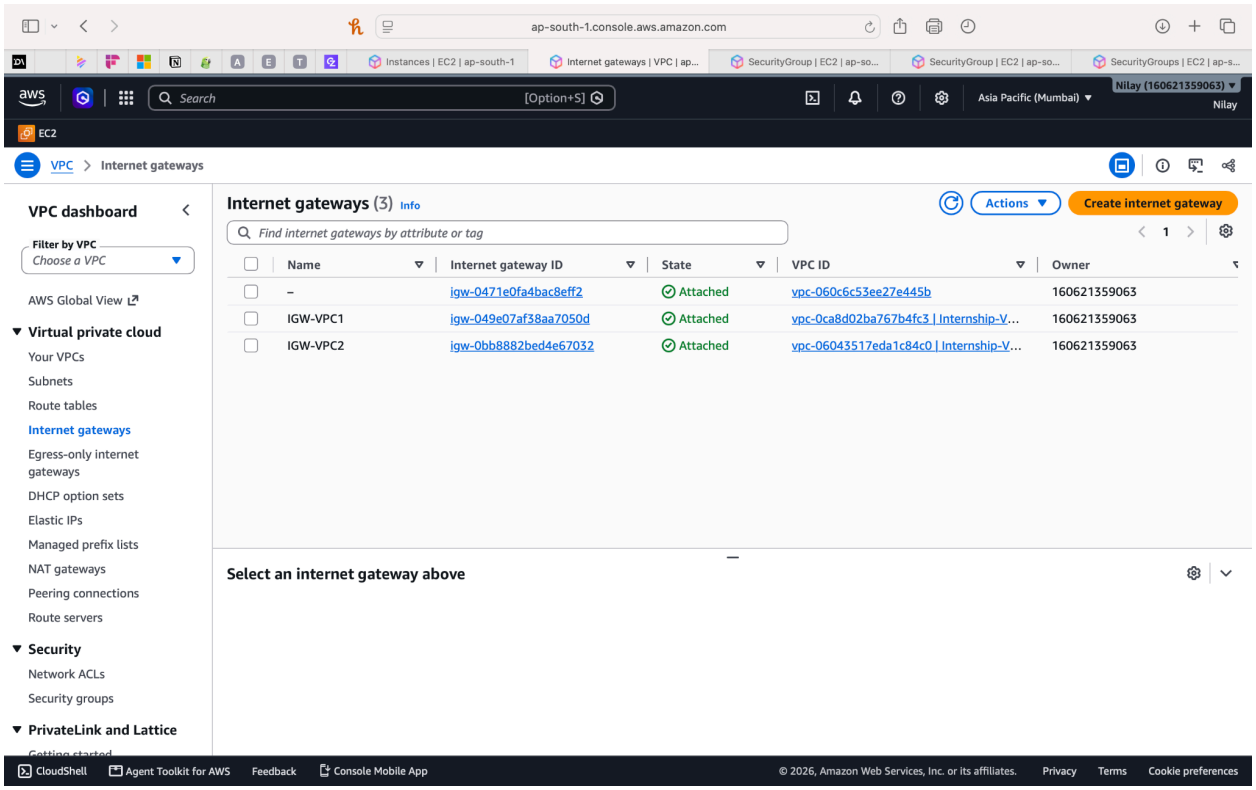
Name: Nilay N Nikam
College: K.K. Wagh Institute of Engineering Education & Research
Course: B.Tech Information Technology
Project: AWS VPC Peering



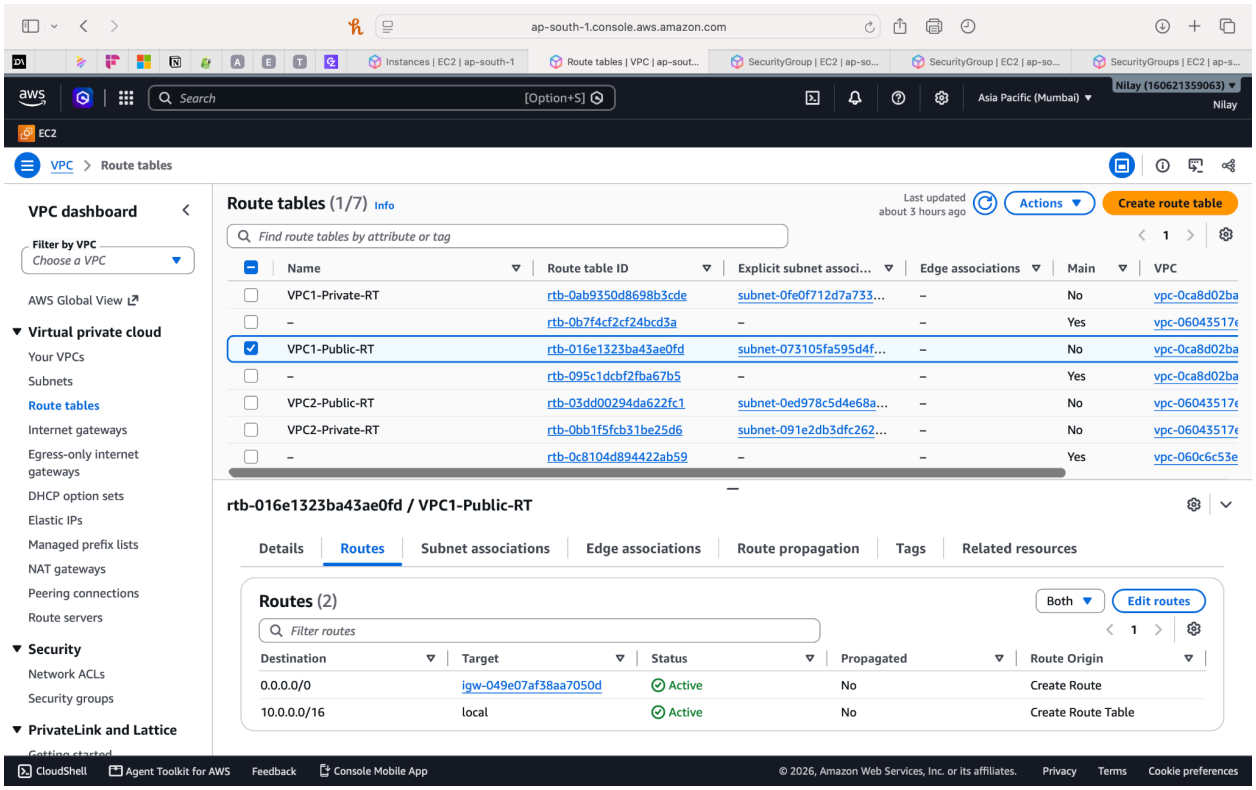
Two custom VPCs, Internship-VPC-1 (10.0.0.0/16) and Internship-VPC-2 (10.200.0.0/16), successfully created in the AWS Mumbai region.



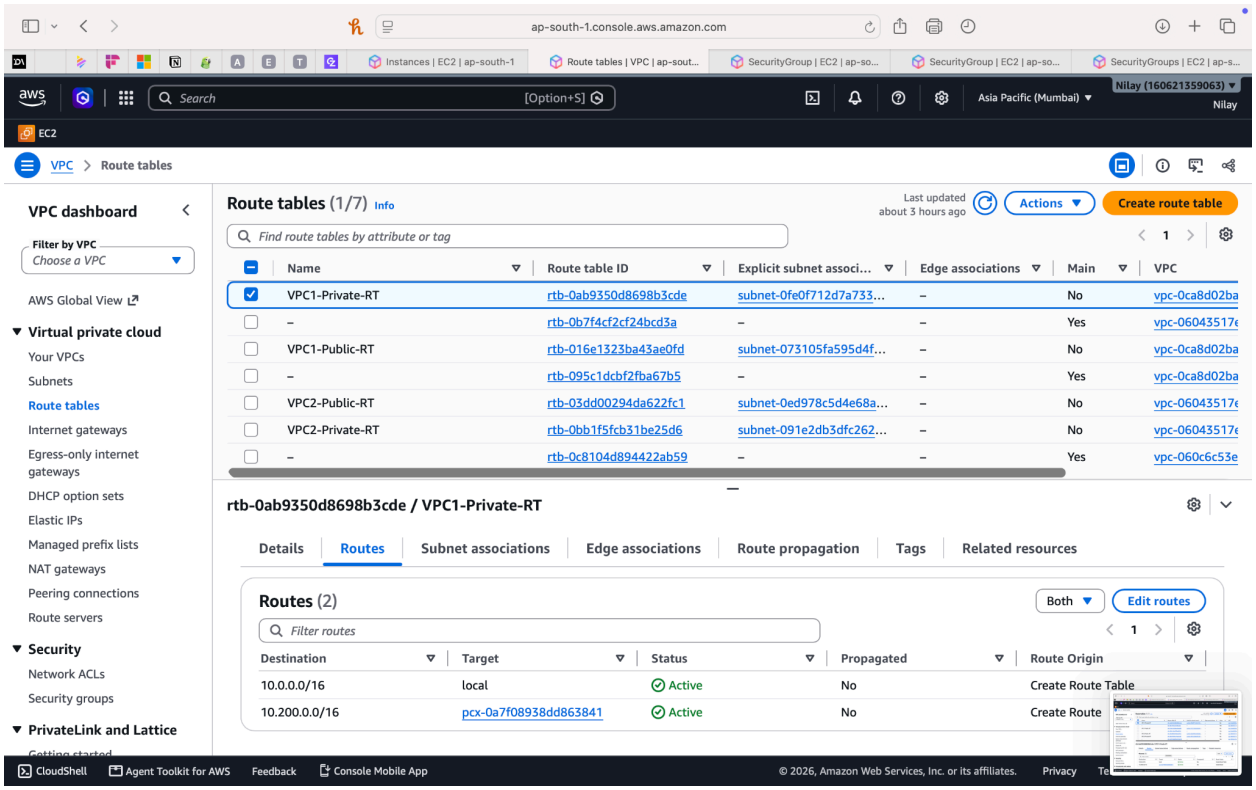
Four subnets configured across the two custom VPCs, including public and private subnets in VPC1 and VPC2.



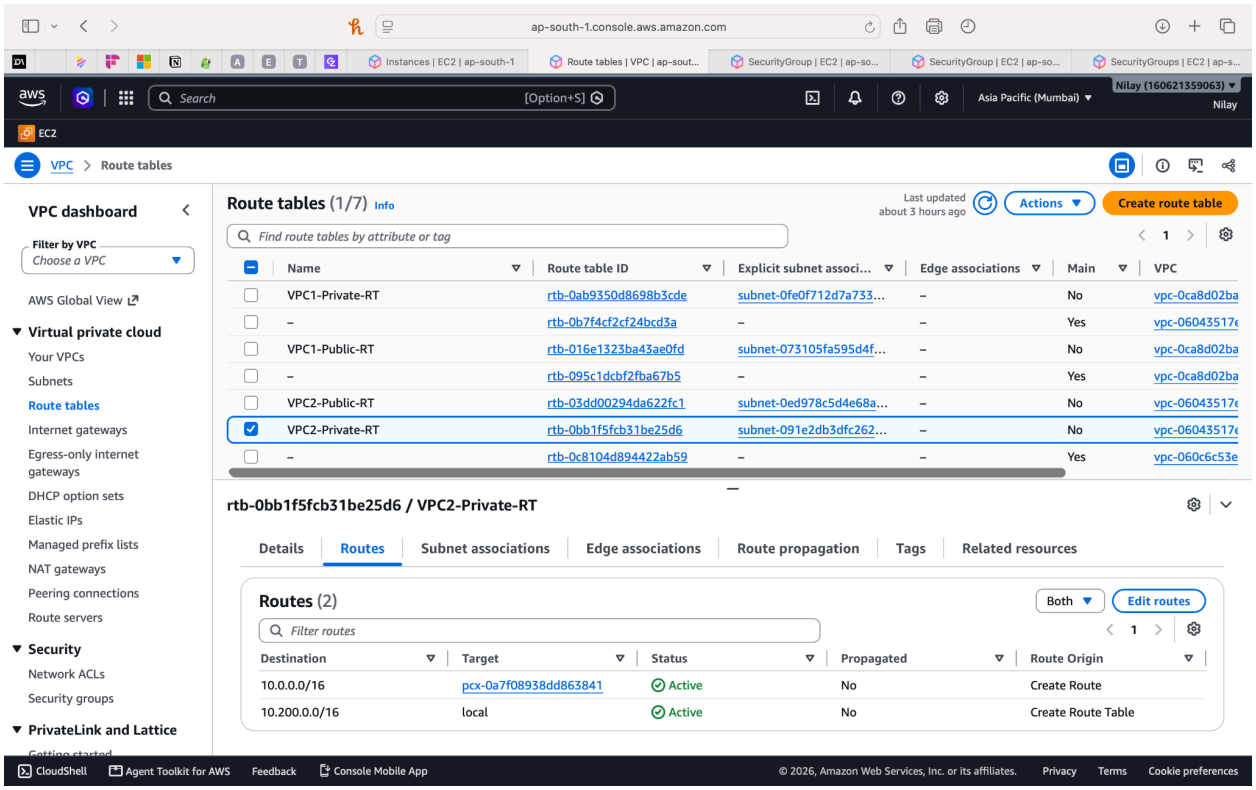
Internet Gateways successfully created and attached to their respective VPCs to provide internet connectivity.



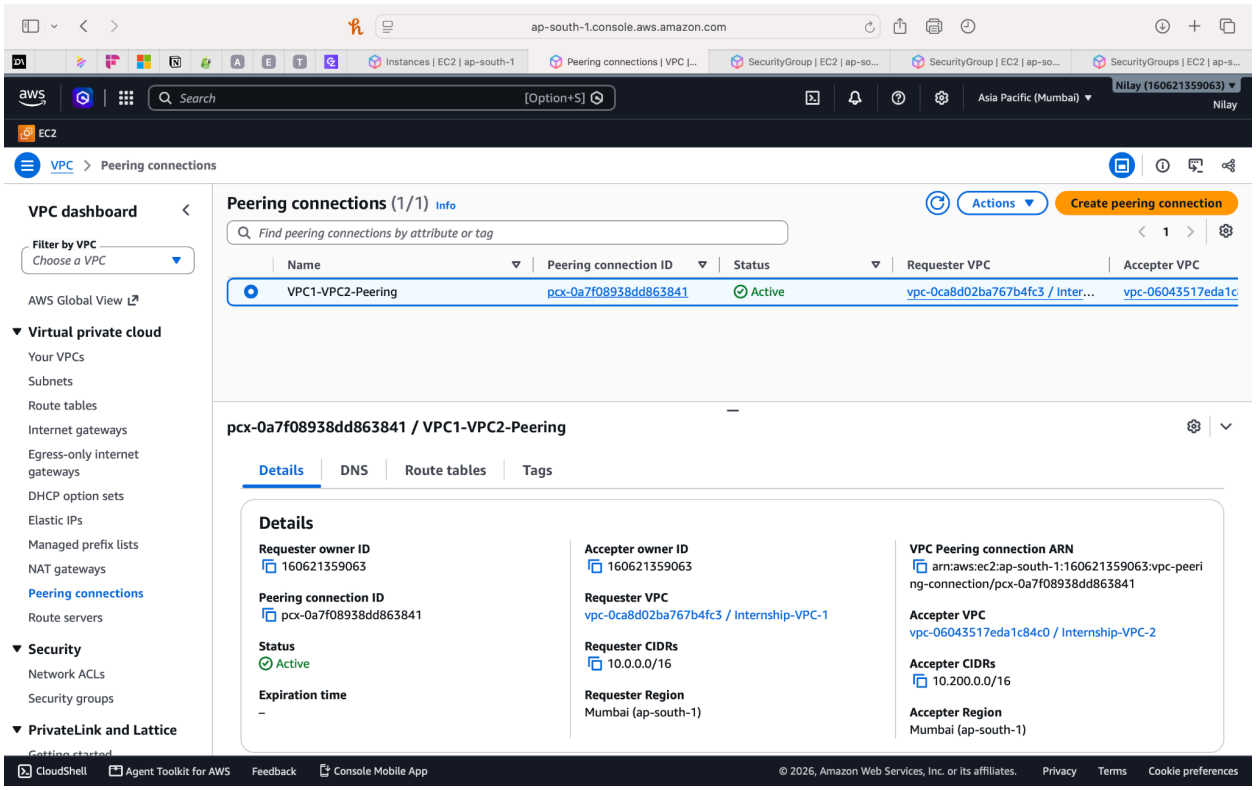
VPC1 public route table configured with a default route to the Internet Gateway and a local route for internal VPC communication.



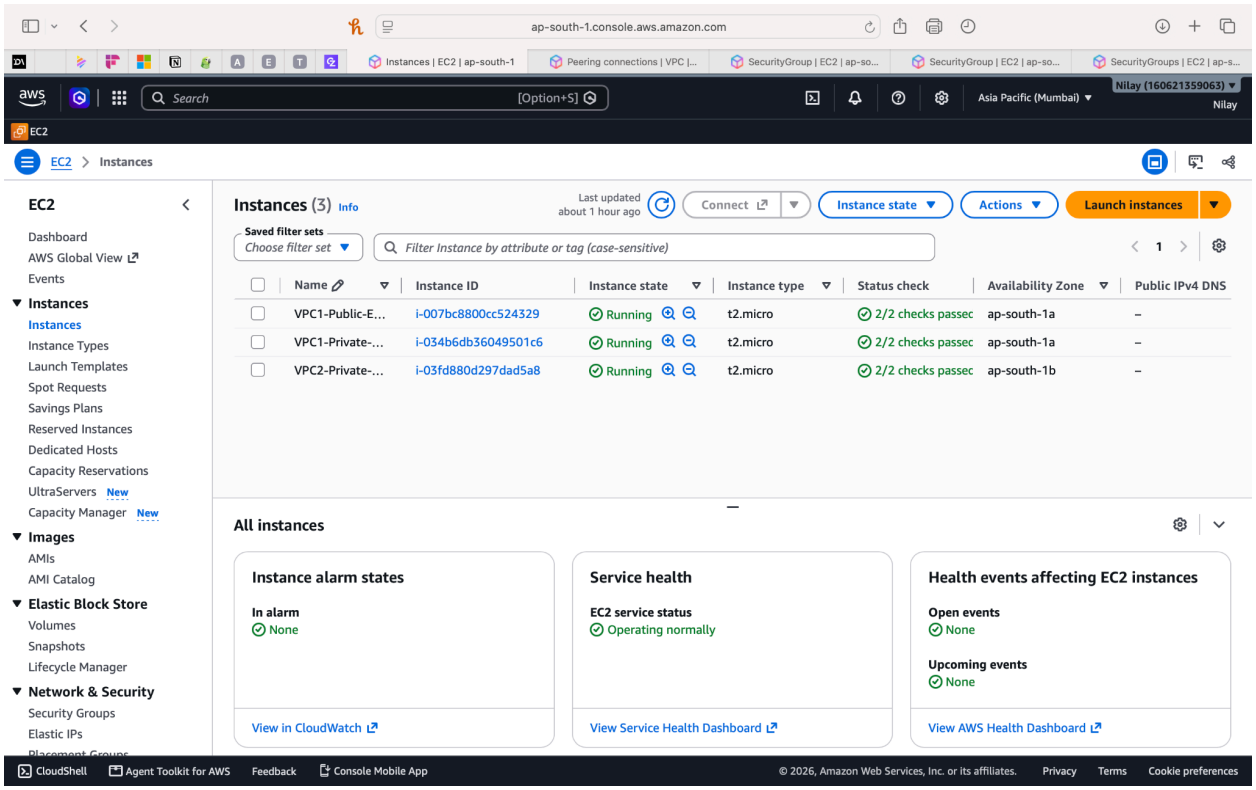
VPC1 Private Route Table configured with a local route and a route to VPC2 through the VPC Peering connection.



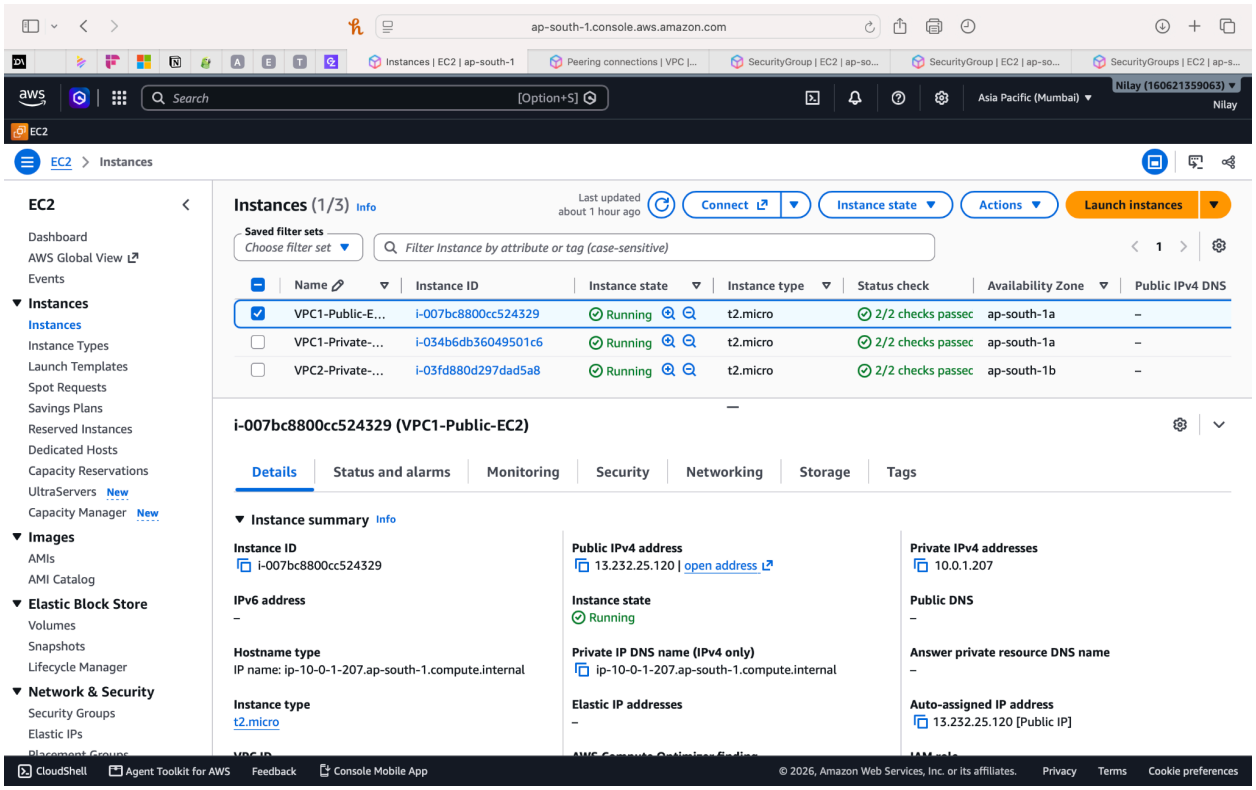
VPC2 Private Route Table showing the local route and the route to VPC1 through the VPC Peering connection.



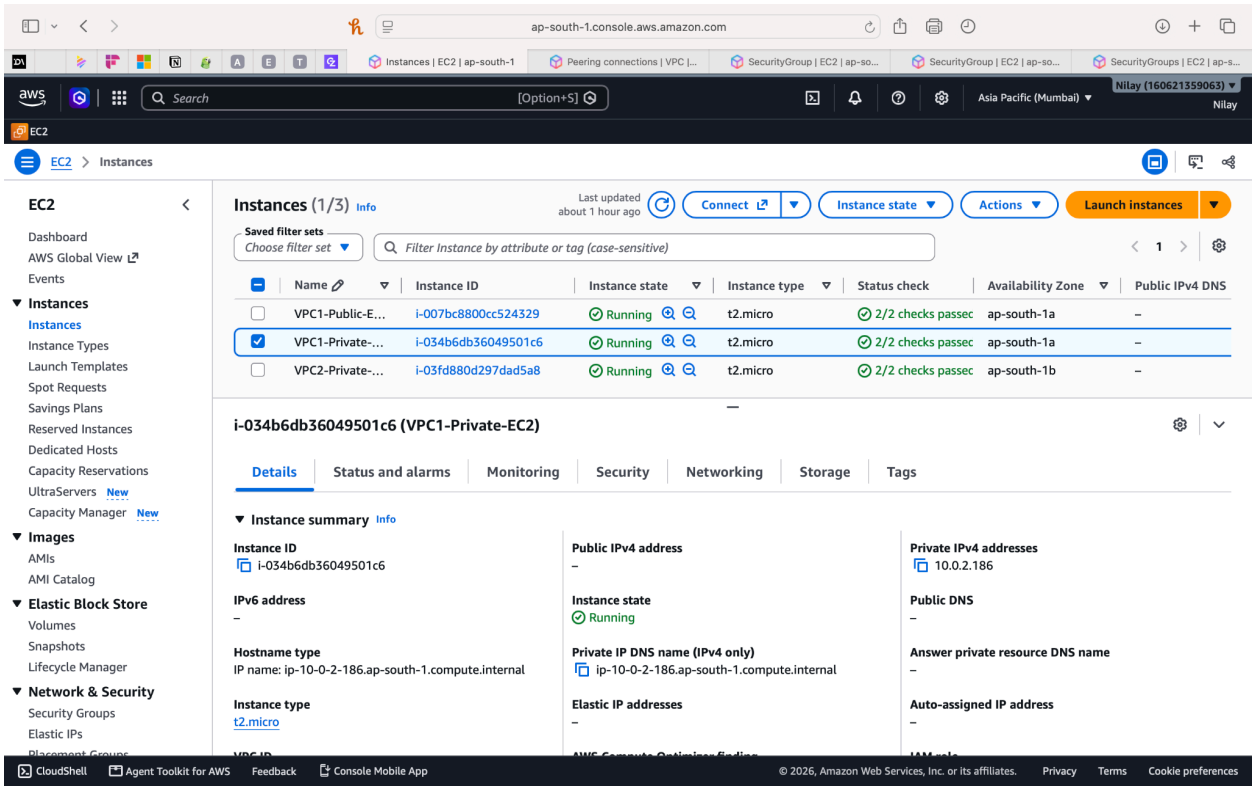
Active VPC Peering connection between Internship-VPC-1 and Internship-VPC-2, showing the requester and accepter VPC details and their CIDR blocks.



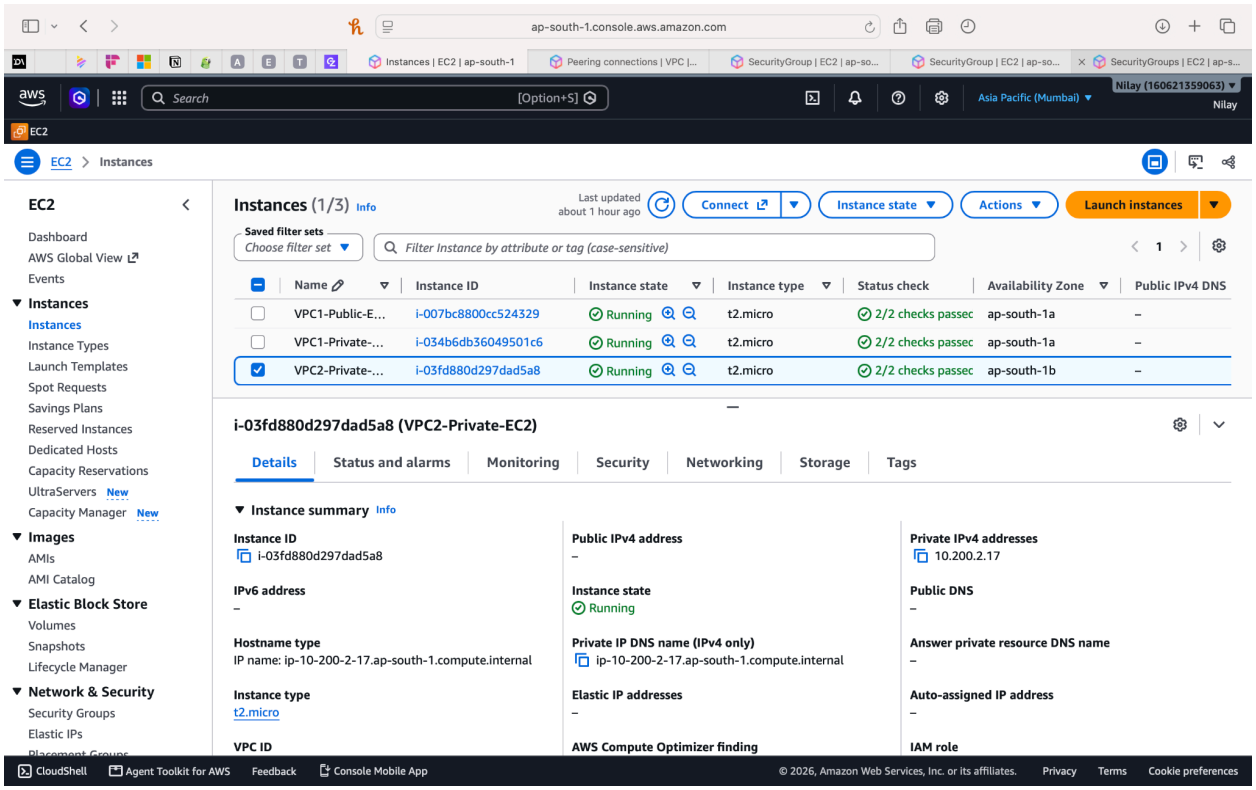
Three EC2 instances deployed across the two VPCs, all in Running state with 2/2 status checks passed.



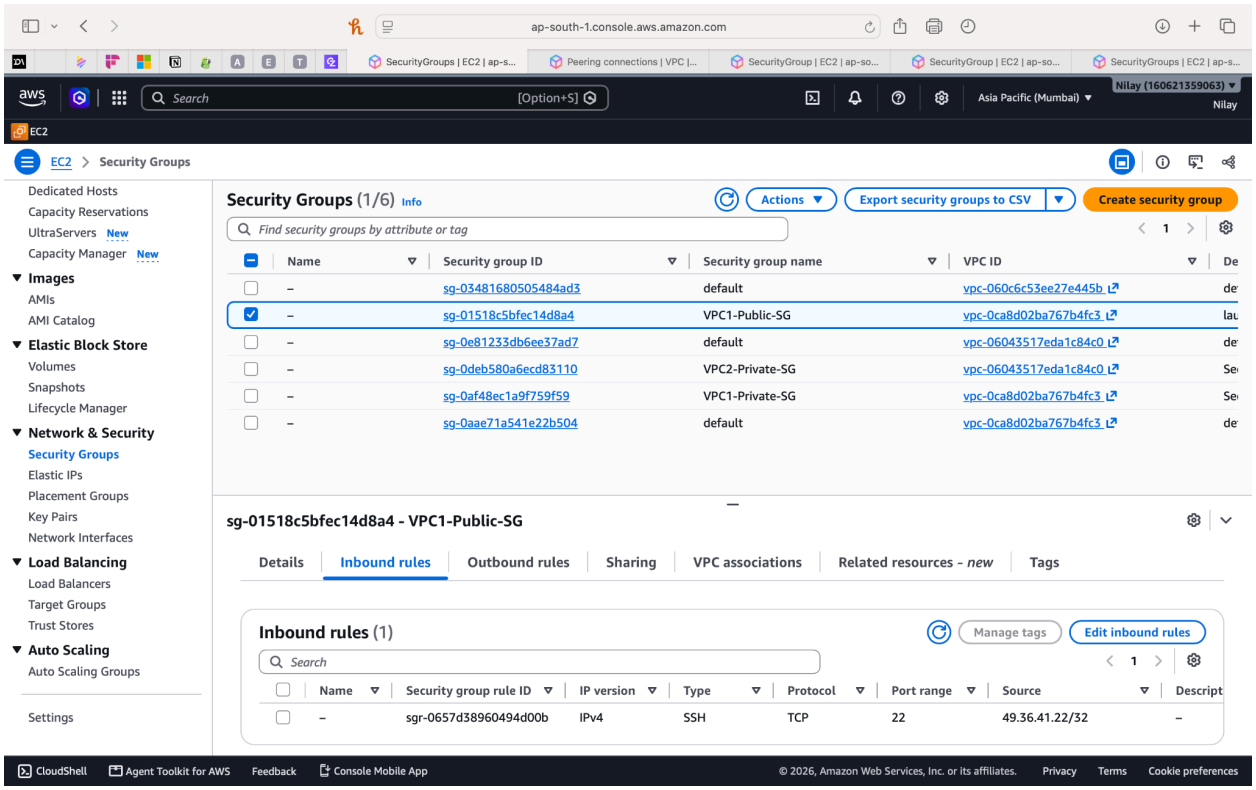
VPC1-Public-EC2 instance details showing its public and private IPv4 addresses.

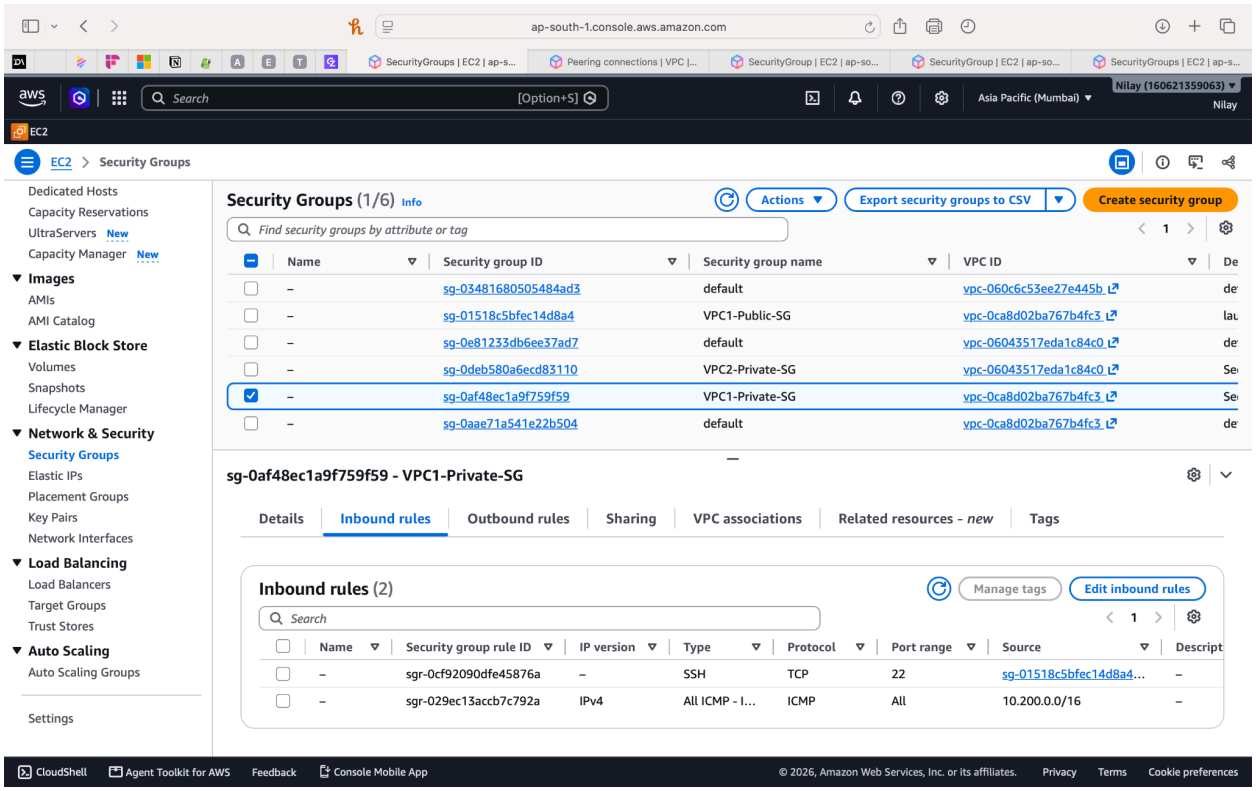


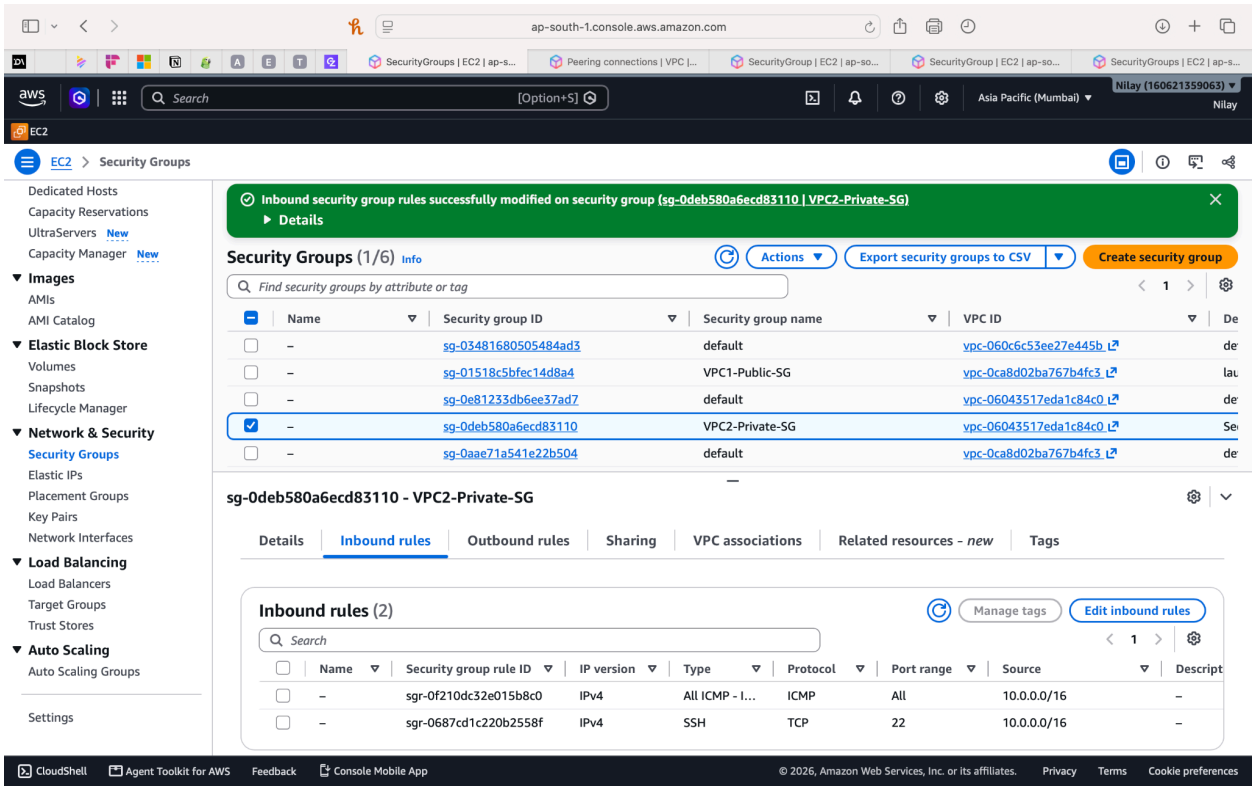
VPC1-Private-EC2 instance details showing its private IPv4 address and confirming that no public IPv4 address is assigned.



VPC2-Private-EC2 instance details showing its private IPv4 address and confirming that no public IPv4 address is assigned.


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Search
[Option+S]
Asia Pacific (Mumbai)
Nilay (160621359063)
Nilay
EC2
EC2 > Security Groups
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UltraServers New
Capacity Manager New
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Security group name
VPC ID
sg-03481680505484ad3 default vpc-060c6c53ee27e445b
sg-01518c5bfec14d8a4 VPC1-Public-SG vpc-0ca8d02ba767b4fc3
sg-0e81233db6ee37ad7 default vpc-06043517eda1c84c0
sg-0deb580a6ecd83110 VPC2-Private-SG vpc-06043517eda1c84c0
sg-0af48ec1a9f759f59 VPC1-Private-SG vpc-0ca8d02ba767b4fc3
sg-0aae71a541e22b504 default vpc-0ca8d02ba767b4fc3
sg-01518c5bfec14d8a4 - VPC1-Public-SG
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sgr-0657d38960494d00b IPv4 SSH TCP 22 49.36.41.22/32
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Nilay
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Security Groups (1/6) Info
Actions
Export security groups to CSV
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Name
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Security group name
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sg-03481680505484ad3 default vpc-060c6c53ee27e445b
sg-01518c5bfec14d8a4 VPC1-Public-SG vpc-0ca8d02ba767b4fc3
sg-0e81233db6ee37ad7 default vpc-06043517eda1c84c0
sg-0deb580a6ecd83110 VPC2-Private-SG vpc-06043517eda1c84c0
sg-0af48ec1a9f759f59 VPC1-Private-SG vpc-0ca8d02ba767b4fc3
sg-0aae71a541e22b504 default vpc-0ca8d02ba767b4fc3
sg-0af48ec1a9f759f59 - VPC1-Private-SG
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IP version
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Port range
Source
sgr-0cf92090dfe45876a – SSH TCP 22 sg-01518c5bfec14d8a4...
sgr-029ec13accb7c792a IPv4 All ICMP - I... ICMP All 10.200.0.0/16
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```
[ec2-user@ip-10-0-1-207 ~]$ ssh -A ec2-user@10.0.2.186
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Oct 31 10:10:37 2026 from 10.0.1.207
[ec2-user@ip-10-0-2-186 ~]$ ssh-add -l
2048 SHA256:nlT6ld314En4lrEfoDm8RllaG63p81U2eiFSTeKpCp8 AWS-Internship-Key.pem (RSA)
[ec2-user@ip-10-0-2-186 ~]$ ssh ec2-user@10.200.2.17
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Oct 31 10:01:44 2026 from 10.0.2.186
[ec2-user@ip-10-200-2-17 ~]$ ping -c 4 10.0.2.186
PING 10.0.2.186 (10.0.2.186) 56(84) bytes of data.
64 bytes from 10.0.2.186: icmp_seq=1 ttl=127 time=0.738 ms
64 bytes from 10.0.2.186: icmp_seq=2 ttl=127 time=1.17 ms
64 bytes from 10.0.2.186: icmp_seq=3 ttl=127 time=0.755 ms
64 bytes from 10.0.2.186: icmp_seq=4 ttl=127 time=1.16 ms

--- 10.0.2.186 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3125ms
rtt min/avg/max/mdev = 0.738/0.954/1.169/0.208 ms
[ec2-user@ip-10-200-2-17 ~]$ ping -c 4 10.0.2.186
PING 10.0.2.186 (10.0.2.186) 56(84) bytes of data.
64 bytes from 10.0.2.186: icmp_seq=1 ttl=127 time=0.713 ms
64 bytes from 10.0.2.186: icmp_seq=2 ttl=127 time=1.26 ms
64 bytes from 10.0.2.186: icmp_seq=3 ttl=127 time=1.09 ms
64 bytes from 10.0.2.186: icmp_seq=4 ttl=127 time=1.06 ms

--- 10.0.2.186 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3043ms
rtt min/avg/max/mdev = 0.713/1.029/1.260/0.198 ms
[ec2-user@ip-10-200-2-17 ~]$ exit
logout
Connection to 10.200.2.17 closed.
[ec2-user@ip-10-0-2-186 ~]$ ping -c 4 10.200.2.17
PING 10.200.2.17 (10.200.2.17) 56(84) bytes of data.
64 bytes from 10.200.2.17: icmp_seq=1 ttl=127 time=0.882 ms
64 bytes from 10.200.2.17: icmp_seq=2 ttl=127 time=1.24 ms
64 bytes from 10.200.2.17: icmp_seq=3 ttl=127 time=0.612 ms
64 bytes from 10.200.2.17: icmp_seq=4 ttl=127 time=0.669 ms

--- 10.200.2.17 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3160ms
rtt min/avg/max/mdev = 0.612/0.849/1.235/0.244 ms
[ec2-user@ip-10-0-2-186 ~]$ Read from remote host 13.232.25.120: Connection reset by peer
Connection to 13.232.25.120 closed.
client_loop: send disconnect: Broken pipe
(base) nilay@Nilays-MacBook-Pro Downloads %
```

Conclusion:

In this project, I successfully set up two VPCs in AWS and connected them using a VPC Peering connection. I configured the required subnets, route tables, security groups, and EC2 instances to make communication between the two networks possible. After completing the setup, I tested the connections using SSH and ping commands. I was able to connect from the public EC2 instance to the private EC2 instance and then access the private instance in the second VPC. I also tested communication in both directions between the two private EC2 instances, and both tests were successful with 0% packet loss.

Working on this project helped me understand how VPCs communicate with each other and gave me practical experience with AWS networking, EC2 instances, routing, and security group configuration.